Accelerated Math II/III Name_____

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HW #4 - Polynomial Long Division

Date_____ Period____

Divide using long division. Write your answer in fraction form.

1) $(5x^4 - 8x^3 - 26x^2 + 16x - 2) \div (x - 3)$

2) $(5x^2 - 23x - 9) \div (x - 5)$

3) $(3x^5 + 5x^4 - 2x^3 + 19x^2 - 30x) \div (x + 3)$

4) $(8x^4 + 8x^3 - 12x^2 - 16x + 6) \div (x + 1)$

5) $(25x^2 - 20x + 1) \div (5x - 2)$

6) $(12x^3 - 29x^2 + 13) \div (3x - 5)$

7) $(6x^3 + 28x^2 + 14x + 1) \div (x^2 + 5x + 4)$

8) $(2x^4 + 15x^3 + 24x^2 - 7x - 15) \div (x^2 + 2x - 3)$

9) $(3x^3 + 8x^2 - 10x - 12) \div (x^2 + 4x + 2)$

10) $(5x^3 - 4x^2 - 11x - 14) \div (x^2 - x - 2)$

Determine if $d(x)$ is a factor of $f(x)$.

11) $f(x) = 3x^3 + 22x^2 + 26x + 12$
$d(x) = x + 6$

12) $f(x) = 12x^3 + 23x^2 - 23x + 1$
$d(x) = 4x + 1$

13) $f(x) = 6x^4 + 4x^3 + 2x^2 + 22x - 28$
$d(x) = 2x + 4$

14) $f(x) = 25x^5 + 5x^4 + 29x^3 + 11x^2 + 25x - 14$
$d(x) = 5x - 2$

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Answers to HW #4 - Polynomial Long Division

1) $5x^3 + 7x^2 - 5x + 1 + \frac{1}{x-3}$

2) $5x + 2 + \frac{1}{x-5}$

3) $3x^4 - 4x^3 + 10x^2 - 11x + 3 - \frac{9}{x+3}$

4) $8x^3 - 12x - 4 + \frac{10}{x+1}$

5) $5x - 2 - \frac{3}{5x-2}$

6) $4x^2 - 3x - 5 - \frac{12}{3x-5}$

7) $6x - 2 + \frac{9}{x^2+5x+4}$

8) $2x^2 + 11x + 8 + \frac{10x+9}{x^2+2x-3}$

9) $3x - 4 - \frac{4}{x^2+4x+2}$

10) $5x + 1 - \frac{12}{x^2-x-2}$

11) Yes

12) No

13) Yes

14) Yes

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